Experiment No. 4

Aim

To develop and execute shell programs using positional parameters, arithmetic operations, conditional statements, and looping constructs in Linux.

Objective

* To understand positional parameters.
* To perform arithmetic calculations in shell scripting.
* To use decision-making statements (if, if-else).
* To implement looping statements (for, while).

A shell script is a text file containing Linux commands executed by the shell interpreter. Shell scripting helps automate repetitive tasks.

1. Positional Parameters

Positional parameters store command-line arguments.

* $0 – Script name
* $1 – First argument
* $2 – Second argument
* $# – Total number of arguments
* $* – All arguments

2. Arithmetic Operations

Arithmetic calculations are performed using:

result=$((a+b))

Operators:

* + Addition

* - Subtraction

* * Multiplication

* / Division

* % Modulus

3. Conditional Statements

Conditional statements execute commands based on conditions.

Syntax:

```
if [ condition ]
then
   commands
else
   commands
fi
```

4.LoopingConstructs

For Loop

```
for i in 1 2 3 4 5
do
   echo $i
```

done

While Loop

```
while[ condition ]
do
   commands
done
```

Algorithm

1. Start.
2. Read two numbers.
3. Perform arithmetic operations.
4. Check which number is greater using if-else.
5. Display numbers from 1 to 5 using a for loop.
6. Stop.

Viva Questions

1. What are positional parameters in Bash?
2. What is the purpose of $0?
3. What does $# represent?
4. How are arithmetic operations performed in shell scripts?
5. What is an if statement?
6. What is the difference between for and while loops?
7. Why is chmod +x used?